Homework 4.5-4.6 #1

Student Name: ______________________

Class Name: **College Algebra FA26 - 014 MW 2:30**

Number of Questions: **12**

Instructor Name: **Mamani Velasco, Ruben**

Question 1 of 12

Use the properties of logarithms to expand the following expression.

$$\log\left(\frac{\sqrt{yz^5}}{x^3}\right)$$

Each logarithm should involve only one variable and should not have any radicals or exponents.

You may assume that all variables are positive.

$$\log\left(\frac{\sqrt{yz^5}}{x^3}\right) = \square$$

Question 2 of 12

Write the expression as a single logarithm.

$$2\log_8 w + 4\left(\log_8 y - 2\log_8 z\right)$$

$\log_{\_\_\_\_\_\_}\left(\_\_\_\_\_\_\right)$

Question 3 of 12

Use the change of base formula to compute $\log_3 \frac{1}{4}$.

Round your answer to the nearest thousandth.

Question 4 of 12

Solve for x.

$$\log_{36} x = \frac{1}{2}$$

Simplify your answer as much as possible.

$x = $ ☐

Question 5 of 12

Solve for x.

$$3 + 5 \ln x = 5$$

Do not round any intermediate computations, and round your answer to the nearest hundredth.

Question 6 of 12

Solve for x.

$$3^{6x} = 27^{3x-1}$$

$x = $ ☐

Question 7 of 12

Suppose that the number of bacteria in a certain population increases according to a *continuous exponential growth model*. A sample of 1900 bacteria selected from this population reached the size of 2103 bacteria in five hours. Find the hourly growth rate parameter.

Note: This is a *continuous* exponential growth model.

Write your answer as a percentage. Do not round any intermediate computations, and round your percentage to the nearest hundredth.

Question 8 of 12

A study is done on the population of a certain fish species in a lake. Suppose that the population size $P(t)$ after t years is given by the following exponential function.

$$P(t) = 530(1.32)^t$$

Find the initial population size. ________
Does the function represent growth or decay? ○ growth ○ decay
By what percent does the population size change each year? ________ $\%$

Question 9 of 12

A radioactive substance decays according to the following function, where y_0 is the initial amount present, and y is the amount present at time t (in days).

$$y = y_0 e^{-0.072t}$$

Find the half-life of this substance. Do not round any intermediate computations, and round your answer to the nearest tenth.

Question 10 of 12

How much should be invested now at an interest rate of 6% per year, compounded *continuously*, to have $\$3000$ in three years?

Do not round any intermediate computations, and round your answer to the nearest cent.

$\$$ ____________

Question 11 of 12

Suppose that the velocity $v(t)$ (in meters per second) of a sky diver falling near the Earth's surface is given by the following exponential function, where time t is the time after diving measured in seconds.

$$v(t) = 53 - 53e^{-0.26t}$$

How many seconds after diving will the sky diver's velocity be 47 meters per second?

Round your answer to the nearest tenth, and do not round any intermediate computations.

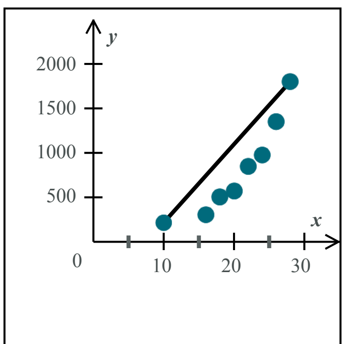
Question 12 of 12

The data points show an item's value y (in dollars) at a time x (in years) after its purchase.

Each figure has the same data points.
However, each figure has a different curve fitting the data.
The equation for each curve is also shown.

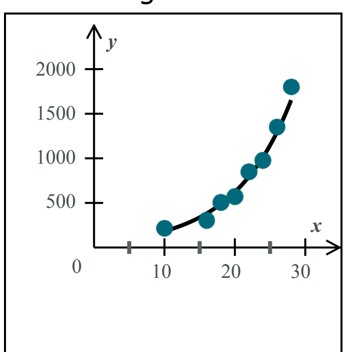
Answer the questions that follow.

Figure 1



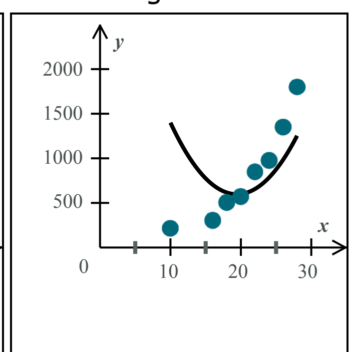
$y = 88.1x - 665$

Figure 2



$y = 54(1.13)^x$

Figure 3



$y = 9x^2 - 350x + 4000$

(a) Which curve fits the data best?

○ Figure 1 ○ Figure 2 ○ Figure 3

(b) Use the equation of the best fitting curve from part (a) to predict the value of the item at a time 12 years after its purchase. Round your answer to the nearest hundredth.

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Question 1 of 12

$$\log\left(\frac{\sqrt{yz^5}}{x^3}\right) = \frac{1}{2}\log y + \frac{5}{2}\log z - 3\log x$$

Question 2 of 12

$$\log_8\left(\frac{w^2 y^4}{z^8}\right)$$

Question 3 of 12

-1.262

Question 4 of 12

$x = 6$

Question 5 of 12

$x = 1.49$

Question 6 of 12

$x = 1$

Question 7 of 12

2.03 %

Question 8 of 12

Find the initial population size.
530
Does the function represent growth or decay?
◉ growth ○ decay
By what percent does the population size change each year?
32%

Question 9 of 12

9.6 days

Question 10 of 12

$2505.81

Question 11 of 12

8.4 seconds

Question 12 of 12

(a) Which curve fits the data best?

○ Figure 1 ◉ Figure 2 ○ Figure 3

(b) Use the equation of the best fitting curve from part (a) to predict the value of the item at a time 12 years after its purchase. Round your answer to the nearest hundredth.

$234.06